

V&V11 — Statistical Assumptions

Normality, independence, equal variance

Shapiro-Wilk: requires $N > 1$; current $N=1$ same-seed disables the test.

→ For $N=30$ branch, walk-distance distributions across replications are checked.

Durbin-Watson: order-by-order autocorrelation of walk_m residuals.

→ Paired-by-order Wilcoxon (used in src/analyze.py) is rank-based and robust to autocorrelation in the differences.

Levene: equal-variance test across policies (used in Welch ANOVA fallback).

→ Collapses under same-seed; documented as a limitation in ASSUMPTIONS.md §23.

